Client Requirements — Voice Khata App

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Role-play brief: written in the voice of a fictional client for practice in requirement gathering. The client, shops and names are invented.

1. Who we are and what hurts

Client: Shri Basaveshwara Kirana Stores (fictional), owner Mallappa Hosamani, speaking for 12 fellow merchants in nearby villages who want the same app.

We need a simple app that replaces our paper khata book, talks to us in Kannada, and tells us at any moment who owes us how much. Today about ₹60,000–1,00,000 of each shop's money sits with customers on credit, and we track it in a notebook.

Our problems, in our words:

- "I write udhaar in a notebook. When it gets wet or a page tears, the money is gone."
- "At rush hour I can't write. I remember later, sometimes I forget."
- "Customers argue: 'I paid you last week.' I have no proof."
- "At night I spend 30–40 minutes adding up the notebook and cash."
- "I don't know which items are finishing until a customer asks and it's gone."
- "My son set up the UPI soundbox, but I still don't know how much came by UPI versus cash today."
- "I read Kannada slowly and English very little. Typing on the phone is hard for me."

2. Business goals and how we will judge success

We will call this project a success if, three months after we start using it, we have stopped using the paper khata and we collect old dues faster.

#	Goal	How we will measure it	Target after 3 months
G1	No lost credit records	Entries missing or disputed	Zero lost entries; disputes settled by showing the history
G2	Faster entry at the counter	Time to record one credit sale	Under 10 seconds by voice

#	Goal	How we will measure it	Target after 3 months
G3	Collect old dues faster	Outstanding credit older than 30 days	Down 25% from the starting figure
G4	Close the day quickly	Time to finish the day's accounts	Under 5 minutes (from 30–40)
G5	No surprise stock-outs	Times a fast-moving item ran out	Warned at least 2 days before
G6	Easy for everyone	Merchants using it daily without help	10 of 12 shops after the first month
G7	No cost to us at the start	Monthly fee	Free during the introductory period; small fee only later, told to us in advance

3. Who will use it

Four kinds of people are involved; only the merchant and helpers use the shop app.

User	Who they are	What they need	Comfort with phones
Merchant (owner)	Shop owner, 35–65 years old, runs the counter	Full control: credit, payments, stock, reports, reminders, settings	Uses WhatsApp and UPI; slow at typing; prefers speaking
Helper	Son, daughter or shop boy	Add credit, payments and bills when the owner is busy; cannot change or delete money entries	Comfortable with smartphones
Customer	Farmers, labourers, families in the village	Only receives balance reminders by SMS or WhatsApp; does not use the app	Many have basic phones; some cannot read

User	Who they are	What they need	Comfort with phones
Your support team	The vendor's team	Help merchants set up, fix problems, see if shops are using the app (no customer names)	Expert

4. What the app must do

Priority: **Must** = we will not use the app without it · **Should** = we want it in the first 3 months · **Could** = nice later.

4.1 Setup and login

ID	Requirement	Priority
FR-01	I can sign up myself from any village or town, without calling anyone	Must
FR-02	I choose my language (Kannada, Hindi, English) on the first screen, and the whole app follows it	Must
FR-03	After the first login, I open the app with a 4-digit PIN or fingerprint	Must
FR-04	I can add my helper with their own PIN and limited rights	Must
FR-05	I can enter old balances from my paper khata quickly, one customer per line	Must
FR-06	The app comes with a ready list of common kirana items so I don't type everything	Should
FR-07	I can use the same account on my phone and on a computer (web)	Should

4.2 Customers and credit (khata)

ID	Requirement	Priority
FR-10	Add a customer with name, village, phone number and optional photo	Must
FR-11	Record credit given: amount, items (optional), date	Must
FR-12	Record payment received: amount, cash or UPI, date; part payments allowed	Must
FR-13	See each customer's balance and full history, newest first	Must
FR-14	See the total money outstanding across all customers	Must
FR-15	List customers who owe more than an amount I choose, or haven't paid for N days	Must
FR-16	Undo a wrong entry within 10 minutes; after that only I (owner) can correct it, with a reason	Must
FR-17	Set a credit limit per customer and warn me when it is crossed	Should
FR-18	Search customers by speaking or typing a part of the name	Must

4.3 Notifications to customers

ID	Requirement	Priority
FR-20	Send one customer their balance by SMS or WhatsApp from my own phone, in the customer's language	Must
FR-21	Select many overdue customers and send reminders one after another	Must
FR-22	Message must be polite and show only shop name and amount due, not item details	Must
FR-23	Optionally send the customer a short receipt after each credit or payment	Should

ID	Requirement	Priority
FR-24	Keep a record of when I last reminded each customer	Should
FR-25	Send a voice-note reminder for customers who cannot read	Could

4.4 Billing and daily transactions

ID	Requirement	Priority
FR-30	Make a quick bill by tapping item pictures or speaking items; pay by cash, UPI or credit (or a mix)	Must
FR-31	Today's screen: total sales, cash, UPI, credit given, old dues collected, cash expected in the drawer	Must
FR-32	Weekly and monthly summary	Should
FR-33	Rough profit for the month (sales minus purchase cost)	Should
FR-34	Show a UPI QR with the exact amount for a customer to pay	Should
FR-35	Print a small bill on a Bluetooth printer	Could
FR-36	GST bill, only for merchants who switch it on	Could

4.5 Stock

ID	Requirement	Priority
FR-40	Keep stock for each item in its unit (kg, litre, packet, piece)	Must
FR-41	Stock reduces automatically when I bill; I can add stock when goods arrive	Must
FR-42	Warn me when an item is below a level I set	Must

ID	Requirement	Priority
FR-43	Tell me how many days of stock are left based on sales	Should
FR-44	Expiry date alerts for packed items	Could
FR-45	Track what I owe my wholesalers	Should

4.6 Reports and data

ID	Requirement	Priority
FR-50	Daily summary sent to my own WhatsApp at closing time if I want	Should
FR-51	Download my full khata as PDF or Excel at any time; the data is mine	Must
FR-52	See who (me or helper) made each entry	Must

5. Voice assistant — what we expect

The voice assistant is the main reason we want this app: I should be able to run my khata by speaking, the way I talk to my customers.

ID	Requirement	Priority
VR-01	One big mic button on every screen; press, speak, done	Must
VR-02	Understand Kannada, Hindi and English, including when I mix them in one sentence	Must
VR-03	Understand our local way of saying numbers ("innooru", "dhai sau", "ondu saavira aidu nooru")	Must
VR-04	Before saving any money entry, read it back to me and show the amount in big numbers; save only after I say yes or tap	Must

ID	Requirement	Priority
VR-05	If two customers have the same name, ask me which one (village, nickname or photo)	Must
VR-06	Answer my questions by speaking back in the language I asked in	Must
VR-07	Work in a noisy shop	Must
VR-08	Basic commands should work without internet	Must
VR-09	If it doesn't understand twice, open the form with whatever it heard, so I can finish by tapping	Must
VR-10	"Cancel the last entry" by voice	Should

Things I will say every day (the app must handle all of these):

- "Ramesh eshtu baaki kodbeku?" — how much does Ramesh owe?
- "Suresh ge 200 saala, akki mattu enne" — ₹200 credit to Suresh for rice and oil
- "Lakshmi 500 kottidaare, PhonePe" — Lakshmi paid ₹500 by UPI
- "Ivattu vyapaara eshtu?" — how much business today?
- "Sakkare eshtu ide?" — how much sugar is left?
- "Yaaru ondu tingalinda kottilla?" — who hasn't paid for a month?
- "Ellarigu reminder kalisu" — send reminders to all of them
- "Ippattu kilo akki banthu" — 20 kg rice arrived

6. How the app must behave (non-functional)

Our network is weak, our phones are cheap, and our khata is our money, so reliability and trust matter more than features.

ID	Area	Requirement
NFR-01	Offline	Everything for daily work (credit, payment, bill, stock, balance) works with no internet; data goes to the cloud by itself when the network returns

ID	Area	Requirement
NFR-02	Phones	Runs smoothly on a ₹7,000–8,000 Android phone with 2–3 GB RAM; app size small enough to download on mobile data
NFR-03	Speed	App opens in under 3 seconds; a saved entry shows instantly; voice reply within 2 seconds when online
NFR-04	Data safety	If my phone breaks or is stolen, I log in on a new phone and get my whole khata back
NFR-05	Privacy	Nobody else, including other merchants, can see my customers or amounts; my data is not sold or shared
NFR-06	Security	PIN lock; helpers cannot delete money entries; every change shows who did it and when
NFR-07	Ease of use	Big buttons, pictures, very few words; I should learn the main work in 15 minutes
NFR-08	Language	Every screen, message and spoken reply available in Kannada, Hindi and English
NFR-09	Battery and data	Should not drain the battery or use much mobile data in a day
NFR-10	Availability	Cloud side available at least 99% of the time; the app itself never stops working because the server is down
NFR-11	Support	Help in Kannada by WhatsApp or phone during shop hours
NFR-12	Scale	Must work for any merchant from any taluk or district; at least 500 customers and 300 items per shop

7. Constraints, out of scope and assumptions

Constraints

- Free for us during the introductory period; any later platform fee must be announced at least 1 month ahead.
- We keep using our existing UPI soundbox; the app must not ask us to change our bank or UPI provider.
- No SMS login codes that we must pay for; we should not pay per message to remind customers.
- Customers will not install anything.

Out of scope for the first version

- Customer login or customer app to check their own balance.
- Online ordering or home delivery.
- Loans, interest calculation or any lending to customers.
- Full accounting (balance sheet, income tax filing).
- GST billing (only later, for merchants who ask).

Assumptions

- Each shop has at least one Android smartphone with WhatsApp.
- The owner or a family member can do the first setup with the vendor's help in under 30 minutes.
- Mobile internet is available at least once a day, even if weak.

Open questions for the vendor

- [ ] What happens to our data if we stop using the app?
- [ ] How much will the platform fee be, and when will it start?
- [ ] Can two phones in the same shop work at the same time?
- [ ] How will you train us — visit, video, or phone call?

8. Acceptance criteria and sign-off

We will accept the first version after a 2-week trial in at least 3 shops, if every check below passes.

#	Test we will do in our shop	Pass when
AC-01	Speak 20 credit and payment entries in Kannada at the counter	At least 17 understood correctly the first time; all 20 saved correctly after read-back
AC-02	Ask the balance of 10 customers by voice	All 10 answers match the khata history
AC-03	Switch off internet for a full day, work normally, switch it on	Every entry appears on the web app the same evening; nothing lost or doubled
AC-04	Log in on a second phone	Full khata appears within 5 minutes
AC-05	Helper tries to delete a payment	App blocks it and asks for the owner's PIN
AC-06	Send reminders to 10 overdue customers	Each gets a correct, polite message in their language
AC-07	Close the day	Today's screen matches our cash and UPI soundbox totals
AC-08	Bill 30 items in a day	Stock levels drop correctly; low-stock warning shows for items below the limit
AC-09	Download the khata	PDF/Excel opens and matches the app
AC-10	A new merchant signs up alone	Ready to use in under 30 minutes with only the in-app guide

Sign-off

Role	Name	Signature	Date
Client (merchant representative)	Mallappa Hosamani (fictional)		
Vendor project lead			